

# DAY 3 - OSI Model & TCP/IP Suite

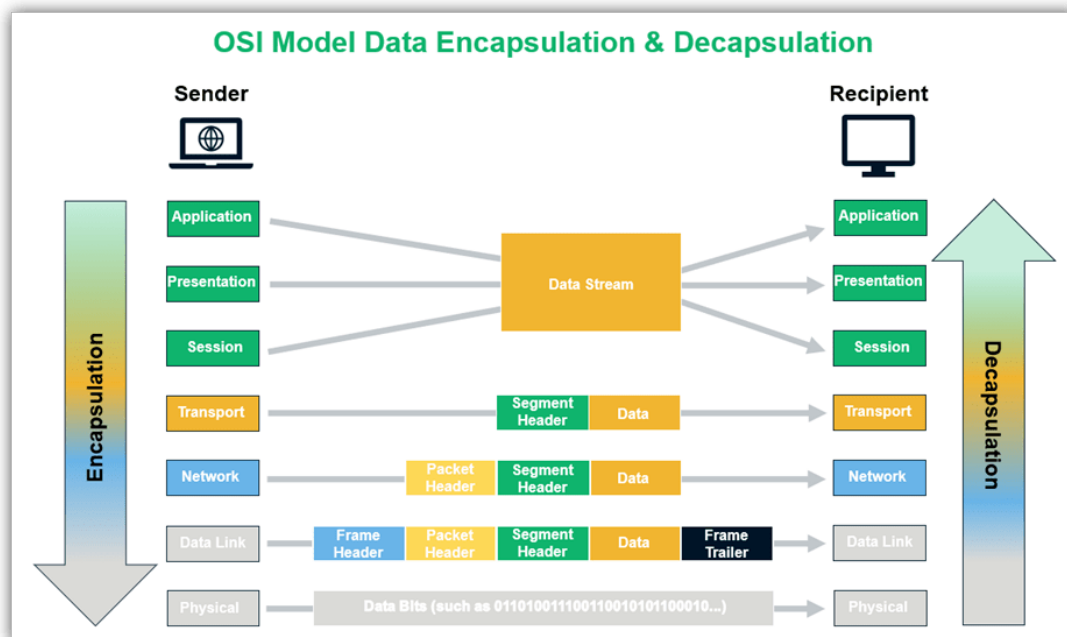
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## OSI Model & TCP/IP Suite

Networking models help categorize and provide a structure for networking **protocols** and **standards**.

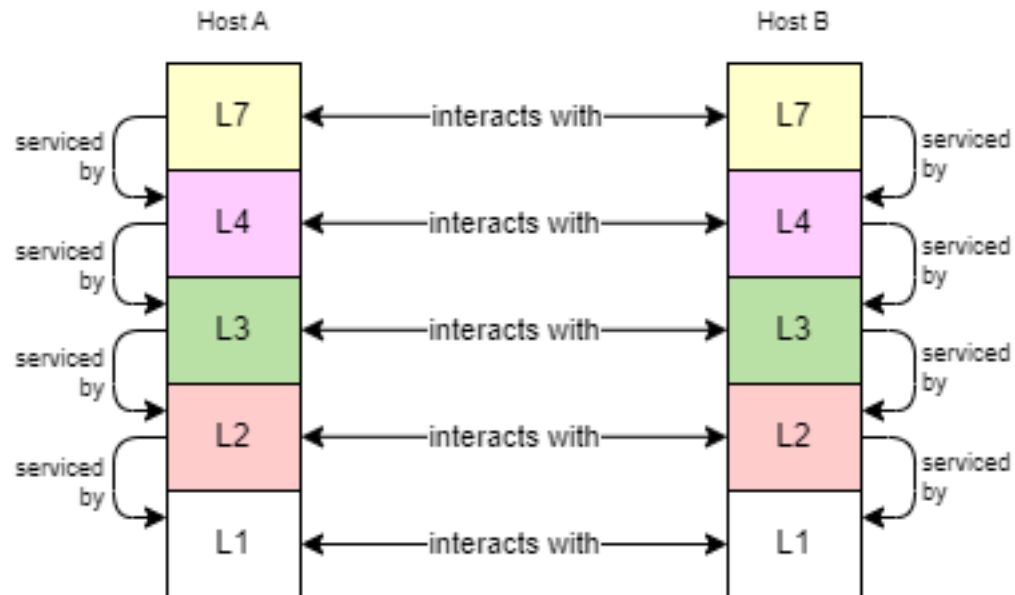
- A **protocol** is a set of rules defining how data should be communicated between devices over a network.
- A **standard** is an agreed-upon specification that describes how a protocol or technology should work.
  - With **vendor-neutral** standards, devices of all types can communicate with each other.

Encapsulation/De-Encapsulation:



- The act of adding header and/or trailer to the data - **Encapsulation**
- The act of removing header and/or trailer to expose data - **De-Encapsulation**

Same Layer vs. Adjacent Layer Interaction:



- **Same Layer Interaction** works on the same layer between 2 Hosts (Network layer to Network layer).
- **Adjacent Layers Interaction** works by having a layer supplying service to the layer directly adjacent to it (eg. Encapsulation, Header attachment)

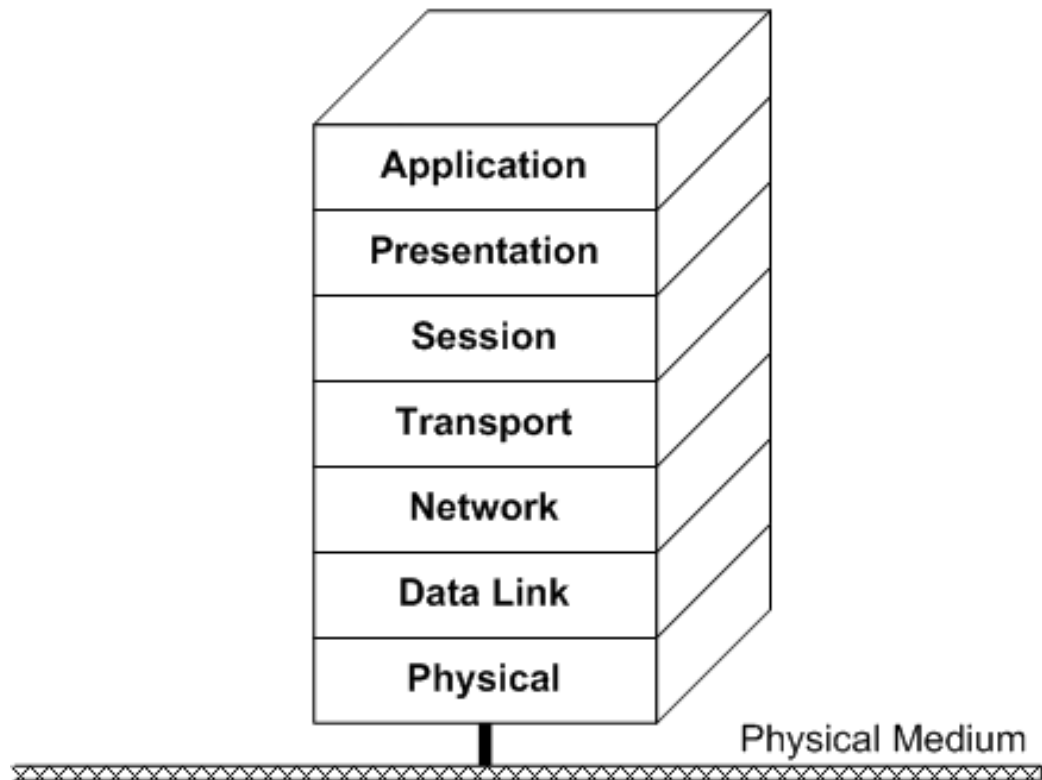
#### Protocol Data Unit (PDU):

The basic unit of data for each layer.

### OSI Model

- Short for **Open Systems Interconnection**
- Created by the International Organization for Standardization (**ISO**)

# The OSI Reference Model



**Layers (From Top to Bottom):**

## **7. Application Layer**

- (a) Closest to the end user.
- (b) Interacts with software applications (Not the software itself)
- (c) **HTTP, HTTPS, DNS** etc.
- (d) Identify communication partner.
- (e) Synchronizing Communication.

## **8. Presentation Layer**

- (a) Translate data into Application Layer's format. (Bits to jpeg, mp4 etc.)
- (b) Encryption & Decryption

## **9. Session Layer:**

- (a) Controls dialogues (Sessions) between communicating hosts.
- (b) Establishes, manages, terminates connections between the **local application** (Browser) and **remote application** (Youtube).

Network Engineers don't really work with the Top-3 Layers.

10. **Transport Layer:**

- (a) **Segments** large data into smaller data and reassemble data back. for easier transmission than 1 large data at once.
- (b) Provides **Host-to-Host** (***Process-to-Process*** is also a popular methodology) communication.
- (c) Add *Layer 4 Header* to the data during encapsulation.
- (d) L4 PDU = **Segment**

11. **Network Layer:**

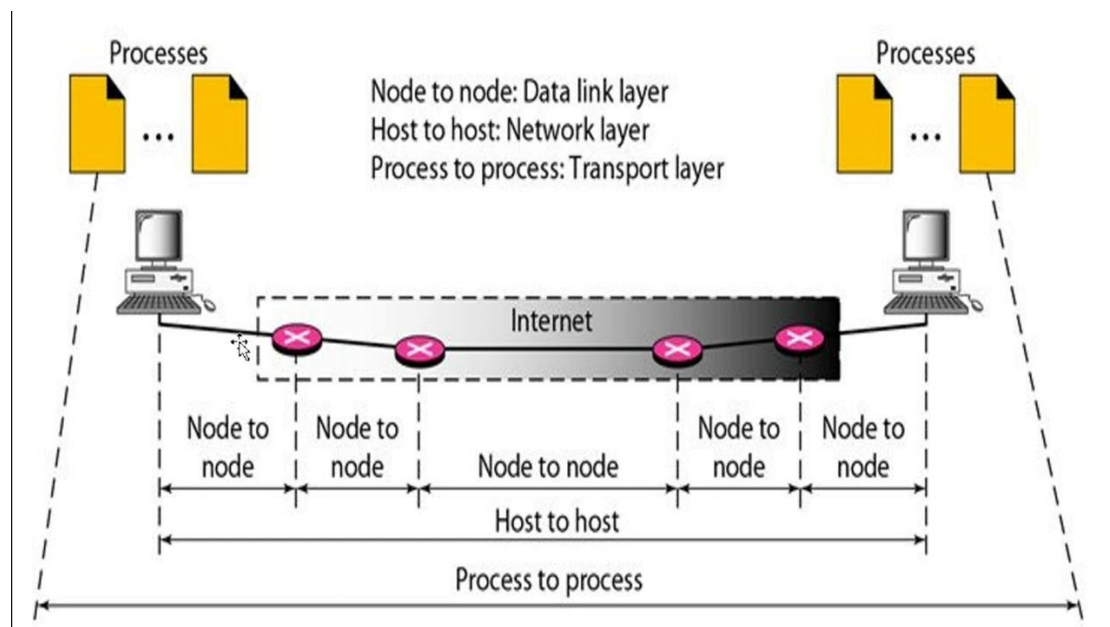
- (a) Provides connectivity between *end-hosts* **on different networks** (eg. outside of the LAN)
- (b) Provide **End-to-End** communication.
- (c) Provide **Logical Addressing (IP Addresses)**
- (d) Layer 3 device: **Router**
- (e) L3 PDU = **Packet**

12. **Data Link Layer:**

- (a) Provide **Hop-to-Hop/Node-to-Node** communication.
- (b) Defines how data is formatted for transmission over a physical medium (eg. copper UTP cables)
- (c) **Detects and corrects Physical Layer errors**
- (d) Provide **Physical Addressing (MAC Addresses)**
- (e) Layer 2 device: **Switch**
- (f) L2 PDU = **Frame**

13. **Physical Layer:**

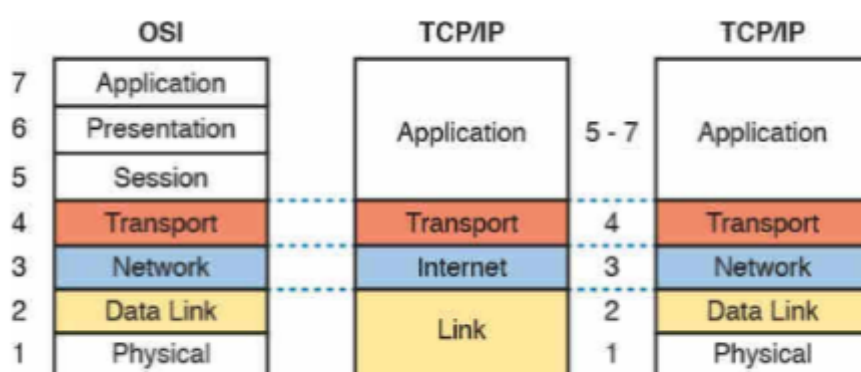
- (a) Define physical characteristics of the medium used to transfer. (eg. Voltage Level, Maximum Transmission Distances, Physical Connectors, Cable etc.)
- (b) Digital bits are converted to electrical (wired) or radio (wireless)
- (c) Information from DAY 2 - Interfaces and Cables are related to this layer.



Process, Host, Node-to-Node visualization

## TCP/IP Suite

- A conceptual model and set of communication protocols used in the Internet and other networks.
- Known as **TCP/IP** because those are two of the foundational protocols of the suite.
- Developed by the United States Department of Defense through *DARPA* (Defense Advanced Research Projects Agency)
- Similar to the **OSI Model**, but with fewer layers.
- This is the model actually in use in modern networks.



(Note that the OSI model still influences how network engineers think and talk about networks. )

- People still mostly refer to the OSI Model when talking in Number (eg.

## Layer 2 Problem = Data Link Layer Problem)

- TCP/IP model might have different layout/naming based on where you look.

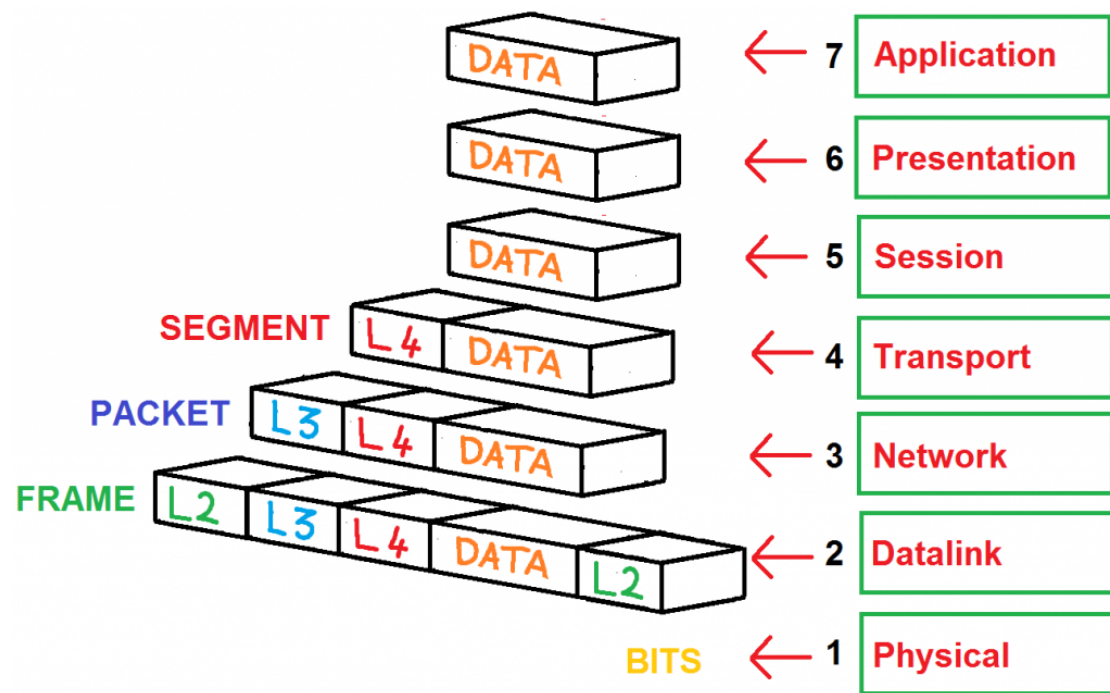
| <a href="#">RFC 1122</a> ,<br>Internet STD 3<br>(1989) | Cisco<br>Academy <sup>[31]</sup> | Kurose, <sup>[32]</sup><br>Forouzan <sup>[33]</sup>          | Comer, <sup>[34]</sup><br>Kozierok <sup>[35]</sup> | Stallings <sup>[36]</sup>    | Tanenbaum <sup>[37]</sup>              | Arpanet<br>Reference Model<br>(RFC 871) | OSI model                              |
|--------------------------------------------------------|----------------------------------|--------------------------------------------------------------|----------------------------------------------------|------------------------------|----------------------------------------|-----------------------------------------|----------------------------------------|
| Four layers                                            | Four layers                      | Five layers                                                  | Four+one<br>layers                                 | Five layers                  | Five layers                            | Three layers                            | Seven<br>layers                        |
| "Internet<br>model"                                    | "Internet<br>model"              | "Five-layer Internet<br>model" or "TCP/IP<br>protocol suite" | "TCP/IP 5-<br>layer<br>reference<br>model"         | "TCP/IP<br>model"            | "TCP/IP 5-layer<br>reference<br>model" | "Arpanet reference<br>model"            | OSI model                              |
| Application                                            | Application                      | Application                                                  | Application                                        | Application                  | Application                            | Application/Process                     | Application<br>Presentation<br>Session |
| Transport                                              | Transport                        | Transport                                                    | Transport                                          | Host-to-host<br>or transport | Transport                              | Host-to-host                            | Transport                              |
| Internet                                               | Internetwork                     | Network                                                      | Internet                                           | Internet                     | Internet                               |                                         | Network                                |
| Link                                                   | Network<br>interface             | Data link                                                    | Data link<br>(Network<br>interface)                | Network<br>access            | Data link                              | Network interface                       | Data link                              |
|                                                        |                                  | Physical                                                     | (Hardware)                                         | Physical                     | Physical                               |                                         | Physical                               |

## Summary:

### Model Comparisons

| OSI Model          | TCP/IP Model      |
|--------------------|-------------------|
| Application Layer  | Application layer |
| Presentation Layer |                   |
| Session Layer      |                   |
| Transport Layer    | Transport Layer   |
| Network Layer      | Internet Layer    |
| Data link layer    | Link Layer        |
| Physical layer     |                   |

## Protocol Data Unit (PDU)



### Layers:

- **Application:**
  - Defines how applications like Mails or Websites will operate.
  - Process-to-Process communication.
  - Identify communication partner.
  - **HTTP, HTTPS, DNS** etc.
- **Presentation:**
  - Format data for Application Layer (image, video, text etc.)
- **Session:**
  - Session establishment, maintenance, and termination
- **Transport:**
  - **Host-to-Host** communication.
  - **Segmentation/Reassembling** of data.
  - **TCP, UDP**
  - Addressing: **Port Numbers**
- **Network:**
  - **End-to-End** communication.
  - **Routing**
  - Devices: **Routers**
  - Addressing: **IP Addresses**
- **Data Link:**
  - **Node-to-Node** communication
  - Format data for physical medium transmission.

- **Detects and corrects Physical Layer errors**
  - Devices: **Switches**
  - Addressing: **MAC Addresses.**
  - The only layer with a Trailer in addition to a Header.
  - **Physical:**
    - Physical medium transmission (eg. UTP Cabling)
    - Electrical or Radio signals.
    - Devices: **Hub, Repeater**
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